

Statistical Modeling (MT-2002)

Date: April 10, 2026

Course Instructors

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Sessional II Exam

Total Time (Hrs): 1

Total Marks: 50

Total Questions: 4

Roll No

Section

Student Signature

Please do not write anything below this line on the first page.

Attempt all the questions

**DO NOT OPEN THE QUESTION BOOK OR START UNTIL
INSTRUCTED.**

Instructions:

1. Attempt on question paper. Read the question carefully, understand the question, and then attempt it.
2. No additional sheet will be provided for rough work.
3. Verify that you have **seven (7)** different printed pages including this title page. There are **four (4)** questions.
4. Calculator sharing is strictly prohibited.
5. Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.
6. Ensure that you do not have any electronic gadget (like mobile phone, smart watch, etc.) with you.
7. Over writing/ cutting cannot be claimed for recheck

Question No.	Q1	Q2	Q3	Q4
Total Marks	10	15	10	15
Obtained Marks				

CLO # 2: Choose appropriate candidate statistical models for a given problem.

Question # 1

[10 = 8 + 2 Marks]

a) Consider the Simple Linear Regression model (SLR) shown below, used to perform a regression task on a dataset:

$$\alpha \sim \langle \text{Distribution} \rangle(\mu_\alpha, \sigma_\alpha)$$

$$\beta \sim \langle \text{Distribution} \rangle(\mu_\beta, \sigma_\beta)$$

$$\epsilon \sim \langle \text{Distribution} \rangle(0, \sigma_\epsilon)$$

$$\mu = \alpha + \beta x_1$$

$$y \sim \langle \text{Distribution} \rangle(\mu, \epsilon)$$

Rewrite the above SLR model into a Multiple Regression Model (MLR) for 3 features using statistical notations. Define the corresponding priors with justifiable distributions and their parameters.

Answer

$$\alpha \sim \text{Normal}(\mu_\alpha, \sigma_\alpha)$$

$$\beta_1 \sim \text{Normal}(\mu_\beta, \sigma_\beta)$$

$$\beta_2 \sim \text{Normal}(\mu_\beta, \sigma_\beta)$$

$$\beta_3 \sim \text{Normal}(\mu_\beta, \sigma_\beta)$$

$$\epsilon \sim \text{Half-Cauchy}(0, \sigma_\epsilon)$$

$$\mu = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

$$y \sim \text{Normal}(\mu, \epsilon)$$

b) By modifying the above SLR model to MLR, are we estimating the parameters of a line, a plane, or a hyperplane?

Answer: By modifying the above SLR model to MLR, we are estimating the parameters of hyperplane.

CLO # 2: Choose appropriate candidate statistical models for a given problem.

Question # 2 If we extend the SLR model to include all three features of the dataset, creating a MLR model, what potential problems could arise? Clearly mention each problem in no more than two lines for each, along with its cause in the data. [15 = 3 * (1 + 2 + 2) Marks]

1. Name: Confounding variable

Problem: The result looks wrong because another hidden factor is affecting both input and output.

Cause: A variable is related to both the predictor and the result.

2. Name: Multicollinearity

Problem: It becomes hard to understand which variable is actually important.

Cause: Some input variables are very similar or strongly related to each other.

3. Name: Masking effect

Problem: An important variable may look unimportant when other variables are added.

Cause: Variables are related in a way that one hides the effect of another.

CLO # 2: Choose appropriate candidate statistical models for a given problem.

Question # 3 [10=3+3+4 Marks]
Consider pooled, unpooled, and hierarchical models for the following grouped data:

$$\text{Data} = \begin{bmatrix} [0, 1, 1, 1, 1, 1] \\ [0, 0, 0, 0, 1, 1] \\ [1, 1, 1, 1, 1, 1] \\ [1, 1, 1, 1, 1, 1] \end{bmatrix}$$

Consider the following hierarchical model:

```
with pm.Model() as model_h1:
    mu = pm.Uniform('mu', 0, 1)
    kappa = pm.Uniform('kappa', 0, 100)

    theta = pm.Beta('theta', alpha=mu*kappa, beta=(1.0-mu)*kappa, shape=4)
    y = pm.Bernoulli('y', p=theta[group_idx], observed=data)

idata_h1 = pm.sample(1000, return_inferencedata=True)
```

1. Identify the model parameters.

Answer: The model parameters are **theta**, which represent the success probabilities for each of the 4 groups in the Bernoulli model.

2. Identify the hyperparameters.

Answer: The hyperparameters are **mu** and **kappa**, which define the Beta prior distribution of **theta**.

3. Out of pooled, unpooled and hierarchical model which one allow information sharing. Justify in one line

Answer: The **hierarchical model** allows information sharing because all group parameters are drawn from a common prior distribution, enabling partial pooling across groups.

CLO # 2: Choose appropriate candidate statistical models for a given problem.

Question # 4

[15 Marks]

1. What does the slope coefficient represent in a simple linear regression model?
 - (a) The variance of residuals
 - (b) The intercept of the regression line
 - (c) The change in the dependent variable for a one-unit change in the independent variable
 - (d) The mean of the dependent variable
2. Consider a hierarchical Gaussian model where group means follow a common distribution. What do the parameters of this common distribution represent?
 - (a) Observed data points
 - (b) Hyperparameters controlling group-level priors
 - (c) Noise in the likelihood function
 - (d) Independent variables in regression
3. The determination coefficient R^2 measures:
 - (a) The variance of residuals
 - (b) The proportion of variance in the dependent variable explained by the independent variables
 - (c) The strength of the relationship between predictors
 - (d) The goodness of fit for logistic models
4. Multicollinearity in multiple linear regression implies:
 - (a) Independent variables are uncorrelated
 - (b) Predictors have constant variance

- (c) Predictors are highly correlated, causing instability in coefficient estimation
 - (d) There is a perfect linear relationship
5. Anscombe's Quartet primarily demonstrates:
- (a) The impact of multicollinearity
 - (b) The importance of data visualization and outlier detection
 - (c) The effect of autocorrelation
 - (d) The challenges of polynomial regression
6. In a Bayesian PyMC model, when should a Bernoulli distribution be used as a likelihood function instead of a Normal distribution?
- (a) When modeling continuous data with symmetric variability around a mean.
 - (b) When modeling binary outcomes, such as success/failure or 0/1 responses.
 - (c) When modeling count data, such as the number of occurrences in a fixed interval.
 - (d) When modeling strictly positive, skewed data like waiting times.
7. Which of the following is a confounding variable?
- (a) A variable correlated with the dependent variable only
 - (b) A variable correlated with both dependent and independent variables
 - (c) A variable that has no influence on the dependent variable
 - (d) An irrelevant variable in the model
8. The masking effect occurs when:
- (a) Residuals have constant variance
 - (b) Outliers distort the model fit
 - (c) The effect of one predictor hides the impact of another
 - (d) Predictors have low correlation
9. In a Bayesian regression approach, a wide posterior distribution for a slope coefficient suggests:
- (a) Multicollinearity
 - (b) High uncertainty in the parameter estimate
 - (c) A strong predictor variable
 - (d) Overfitting of the model
10. Why use synthetic data in linear regression modeling?
- (a) To estimate multicollinearity
 - (b) To validate model assumptions under controlled scenarios
 - (c) To optimize hyperparameters
 - (d) To minimize residual variance

For remaining MCQs Consider the following PyMC models

- (a) with pm.Model() as model:
 alpha = pm.Normal("alpha", mu=0, sigma=1)
 beta = pm.Normal("beta", mu=0, sigma=1, shape=1)
 epsilon = pm.HalfCauchy("epsilon", beta=1)
 mu = alpha + beta[1] * X
 Y_obs = pm.Normal("Y_obs", mu=mu, sigma=epsilon, observed=Y)
- (b) with pm.Model() as model:
 alpha = pm.Normal("alpha", mu=0, sigma=1)
 beta = pm.Normal("beta", mu=0, sigma=1, shape=2)
 epsilon = pm.HalfCauchy("epsilon", beta=1)
 mu = alpha + beta[0] * X1 + beta[1] * X2
 Y_obs = pm.Normal("Y_obs", mu=mu, sigma=epsilon, observed=Y)
- (c) with pm.Model() as model:
 alpha = pm.Normal("alpha", mu=0, sigma=1)
 beta = pm.Normal("beta", mu=0, sigma=1, shape=2)
 epsilon = pm.HalfCauchy("epsilon", beta=1)
 mu = alpha + beta[0] * X + beta[1] * X**2
 Y_obs = pm.Normal("Y_obs", mu=mu, sigma=epsilon, observed=Y)
- (d) with pm.Model() as model:
 alpha = pm.Normal("alpha", mu=0, sigma=1)
 beta = pm.Normal("beta", mu=0, sigma=1, shape=1)
 v = pm.Exponential("v", lam=1)
 mu = alpha + beta[0] * X
 Y_obs = pm.StudentT("Y_obs", mu=mu, sigma=1, nu=v, observed=Y)

11. Which of the above mentioned PyMC models correctly represents polynomial greater than degree 1 regression?

Options: a, b, c, d

12. Which one of the mentioned above PyMC models represents a simple linear regression model?

Options: a, b, c, d

13. Which one of the above mentioned PyMC models represents a multiple linear regression model?

Options: a, b, c, d

14. Out of above in which model we expect the three problems?

Options: a, b, c, d

15. Out of the above model, what you think which one will not execute or generate error?

Options: a, b, c, d

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Instruction for filling the sheet 1. This sheet should not be folded or crushed 2. Use only blue/black ball pen or 2HB pencil 3. Circle should darkened completely and properly 4. Erase marked circle completely for deselection	WRONG METHOD  CORRECT METHOD 
NAME : _____	

Roll No 0 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 1 ☐ ☐ ☐ ☐ ☐ ☐ ☐ 2 ☐ ☐ ☐ ☐ ☐ ☐ ☐ 3 ☐ ☐ ☐ ☐ ☐ ☐ ☐ 4 ☐ ☐ ☐ ☐ ☐ ☐ ☐ 5 ☐ ☐ ☐ ☐ ☐ ☐ ☐ 6 ☐ ☐ ☐ ☐ ☐ ☐ ☐ 7 ☐ ☐ ☐ ☐ ☐ ☐ ☐ 8 ☐ ☐ ☐ ☐ ☐ ☐ ☐ 9 ☐ ☐ ☐ ☐ ☐ ☐ ☐ MT2002 Sessional II A B C D 1 ☐ ☐ ☐ ☐	A B C D 2 ☐ ☐ ☐ ☐ 3 ☐ ☐ ☐ ☐ 4 ☐ ☐ ☐ ☐ 5 ☐ ☐ ☐ ☐ A B C D 6 ☐ ☐ ☐ ☐ 7 ☐ ☐ ☐ ☐ 8 ☐ ☐ ☐ ☐ 9 ☐ ☐ ☐ ☐ 10 ☐ ☐ ☐ ☐ A B C D 11 ☐ ☐ ☐ ☐ 12 ☐ ☐ ☐ ☐ 13 ☐ ☐ ☐ ☐ 14 ☐ ☐ ☐ ☐	A B C D 15 ☐ ☐ ☐ ☐
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Answer key

Q No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Correct	C	B	B	C	B	B	B	C	B	B	C	A	B	B	A